

EXECUTIVE SUMMARY

USC BDHSC Student Case Competition – Team HA

Research Question

How can predicted treatment outcomes and patient preferences be transparently integrated into a clinical decision-support framework that preserves interpretability, trust, and supports shared decision making?

Background and Motivation

Decisions between surgery and conservative care for musculoskeletal conditions require balancing multiple competing outcomes, including long-term pain relief, short-term treatment-related pain, functional improvement, rehabilitation time, and out-of-pocket cost. These outcomes vary substantially across patients, and their relative importance differs based on individual values and circumstances.

While causal and machine learning models can estimate patient-specific treatment effects, prediction alone is insufficient for real-world clinical decisions. Clinicians must explain trade-offs, acknowledge uncertainty, and incorporate patient values during treatment discussions. This challenge requires not only accurate outcome predictions, but also a transparent and interpretable decision framework that can be used directly within clinical encounters to support shared decision making (SDM).

Predicted Outcomes and Data Inputs

This clinical decision-support (CDS) framework integrates two key inputs developed across the competition. First, patient-specific treatment effects from Challenge 1 estimate how surgery versus conservative care is expected to affect each patient across five outcome domains. These predictions represent individualized counterfactual comparisons rather than population averages.

Second, Dataset B provides patient-reported preference weights (WP1–WP5), collected through a 100-chip allocation survey to capture how individuals prioritize outcomes. Because patients did not always allocate exactly 100 chips, all preference weights were normalized to sum to 100. This preserves relative priorities while ensuring fair comparability across patients.

Outcome Normalization and Decision Framework

The five outcomes are measured on different scales (scores, weeks, and dollars) and cannot be directly aggregated. Outcome directions were first aligned so that positive values consistently represent benefit from surgery. Outcomes were then converted to a common dollar-value scale using health-system willingness-to-pay estimates for pain reduction and functional improvement, along with transparent assumptions for rehabilitation time and direct costs.

A multi-attribute, preference-weighted utility model combines normalized patient preferences with scaled treatment effects to compute a patient-specific utility difference between surgery and conservative care. Treatment recommendations are based on the sign and magnitude of this individualized utility comparison rather than population-level averages. To avoid overconfident automation, a decision margin is applied to identify borderline cases where differences are small or unstable, explicitly flagging these cases for clinician–patient discussion.

Predicted Recommendation Patterns and Robustness

Applying the framework to Dataset B reveals clinically plausible population-level patterns. Approximately 72% of patients are recommended conservative care, while 28% are recommended surgery, indicating that surgery is not a default recommendation but is reserved for patients expected to benefit meaningfully. Most decisions lie near the decision threshold, resulting in a large proportion of borderline cases, while high-confidence recommendations are relatively rare but decisive when present.

Sensitivity analyses further assess robustness by evaluating how recommendations change under plausible shifts in patient preferences. Stable recommendations are labeled as robust, while preference-sensitive cases are identified to highlight decisions that depend strongly on patient values.

Clinical Integration and Conclusion

The CDS prototype presents recommendations alongside outcome-specific contributions, preference visualizations, confidence levels, and sensitivity indicators. Rather than delivering a single “answer,” the tool is designed to surface trade-offs and uncertainty, prompting meaningful clinician–patient discussion during treatment visits.

By integrating individualized treatment effects with normalized patient preferences on a common value scale, this CDS framework delivers transparent, patient-centered recommendations while explicitly supporting shared decision making. The system is designed to support and cannot replace clinical judgment, ensuring that treatment decisions reflect both clinical evidence and what matters most to each patient.